

Emily Hoskins

WA Bee Atlas Collection and Identification Report - 2024

1 Your 2024 Collections

Emily Hoskins caught 11 bees across 1 county from April 23, 2024 to June 05, 2024 representing 4 unique taxa, including 3 unique species.

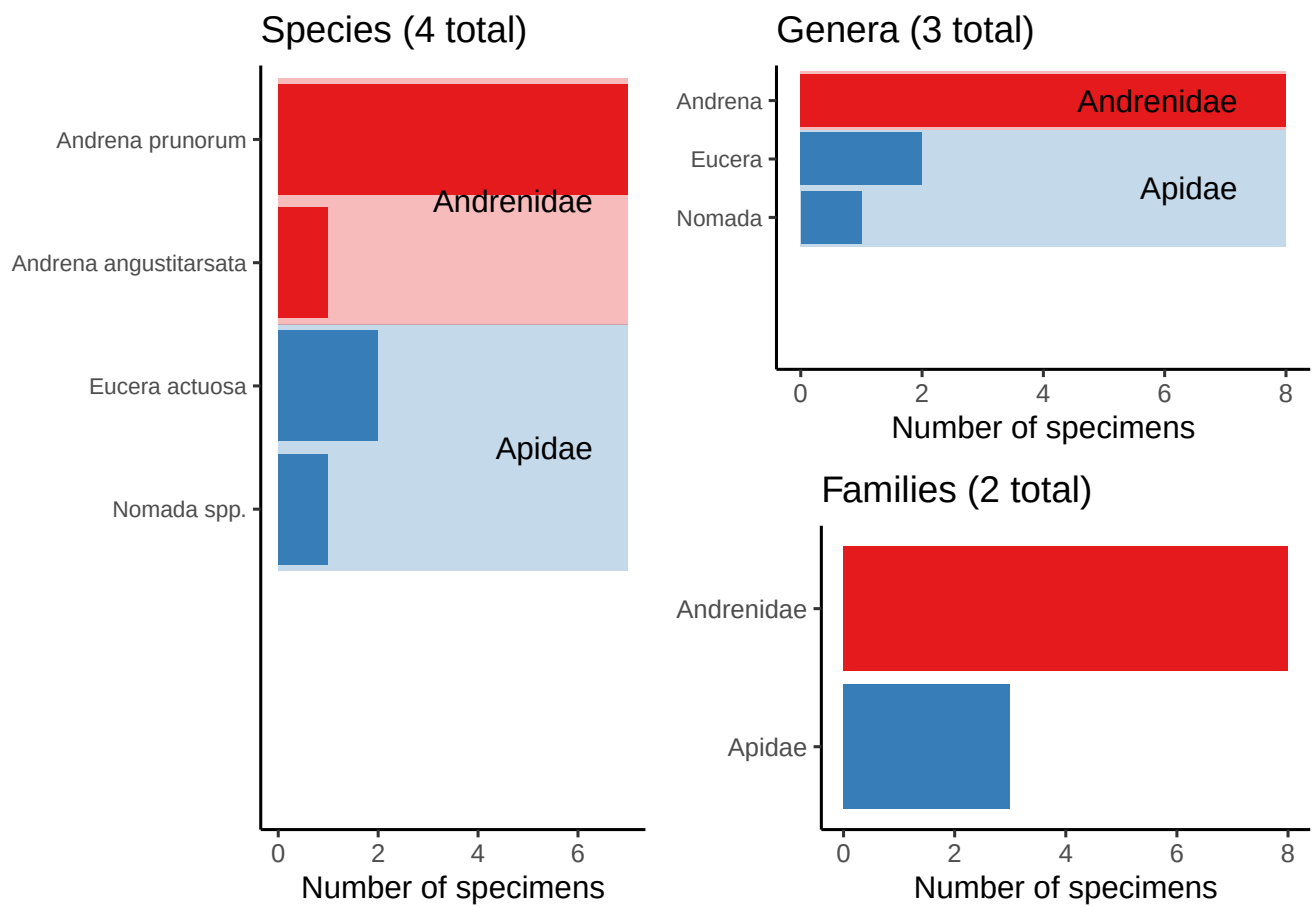


Figure 1: Bees caught by Emily Hoskins, broken down by species, genus, and family.

2 All Your Collections

Emily Hoskins caught 11 bees across 1 county from April 23, 2024 to June 05, 2024, representing 4 unique taxa, including 3 unique species.

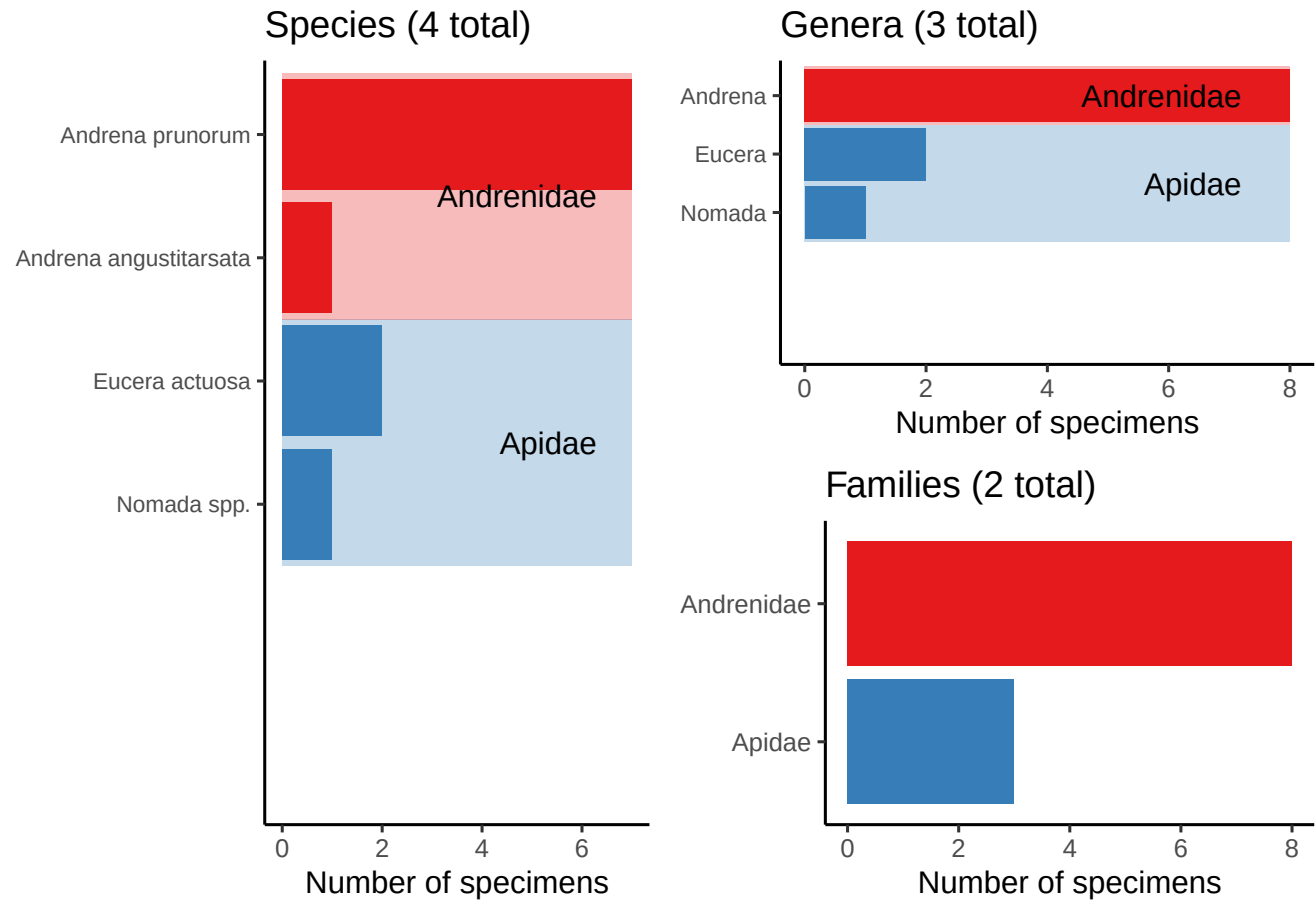


Figure 2: Bees caught by Emily Hoskins, broken down by species, genus, and family.

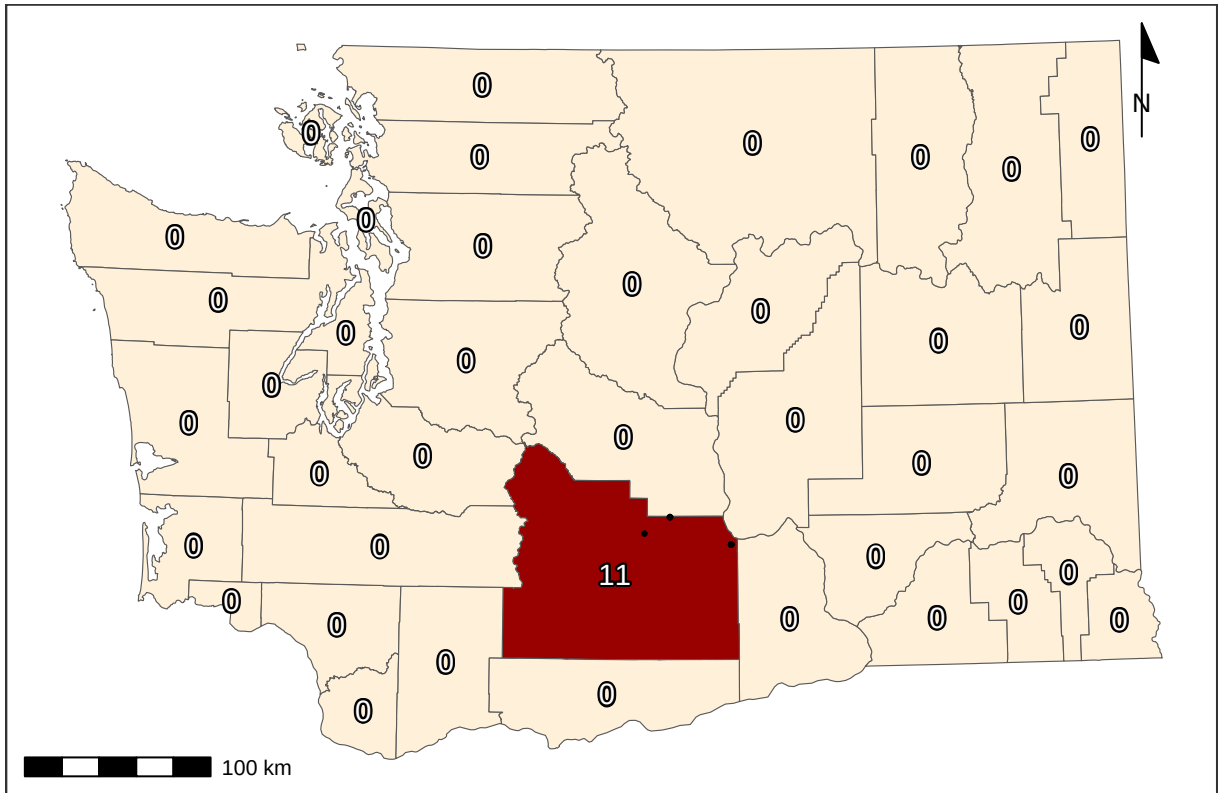


Figure 3: Bee catch locations for Emily Hoskins (within Washington), along with total catches per county.

3 Total Catches

Volunteers from the Washington Bee Atlas project caught 10006 bees across 35 counties from March 10, 2024 to November 14, 2024, representing 354 unique species and 43 unique genera. The **Nimble Net Kudos** (most specimens collected) goes to Karla Salp, Karen Wright, and Ingrid Carmean, who caught a total of 1689, 1005, and 957 specimens. The *positive* kind of **Darwin Award** (most species collected) goes to Karla Salp, Karen Wright, and Ellery Newcomer, who caught a total of 220, 167, and 153 unique species. Well done!

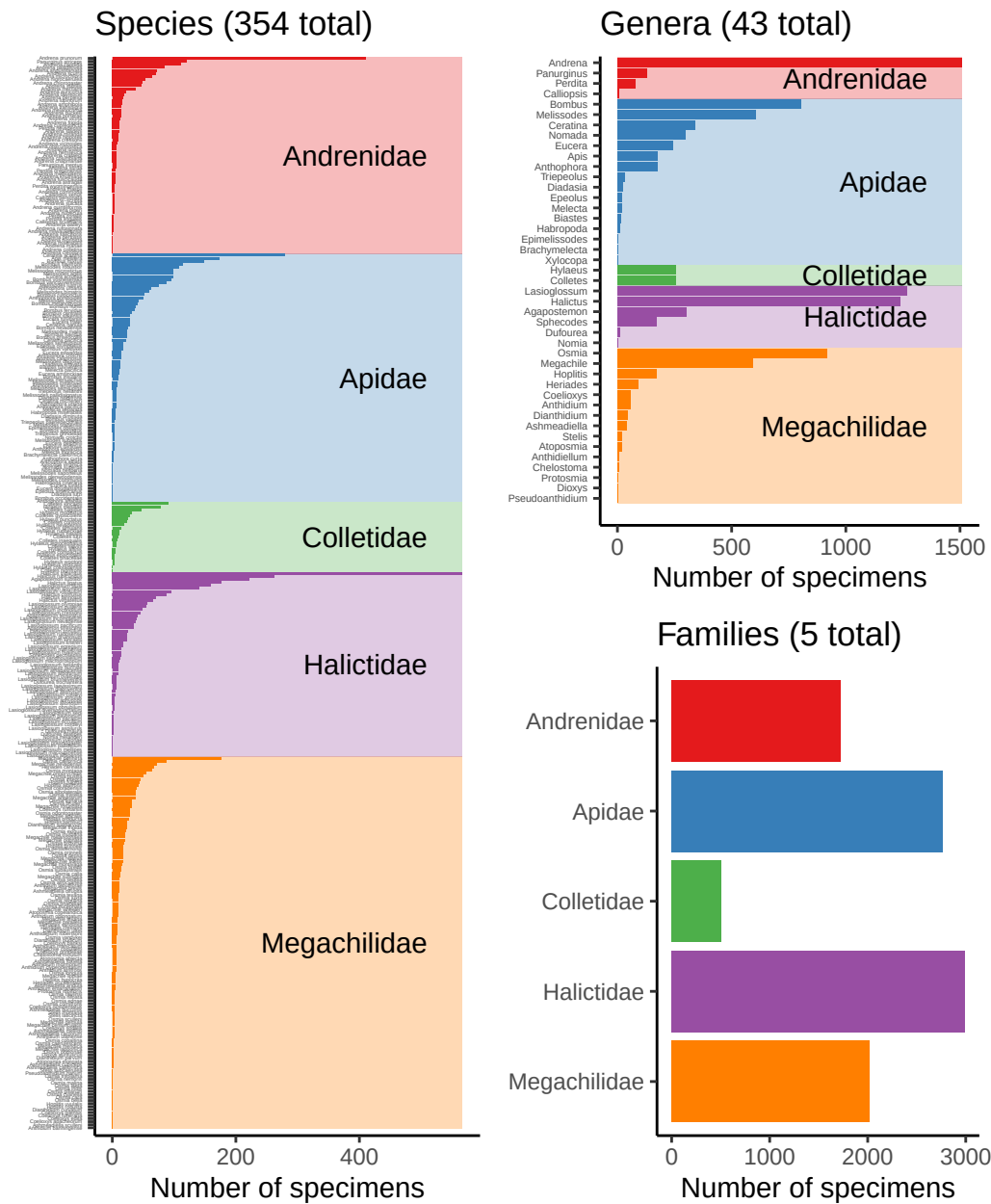


Figure 4: Bees caught by all volunteers, broken down by species, genus, and family.

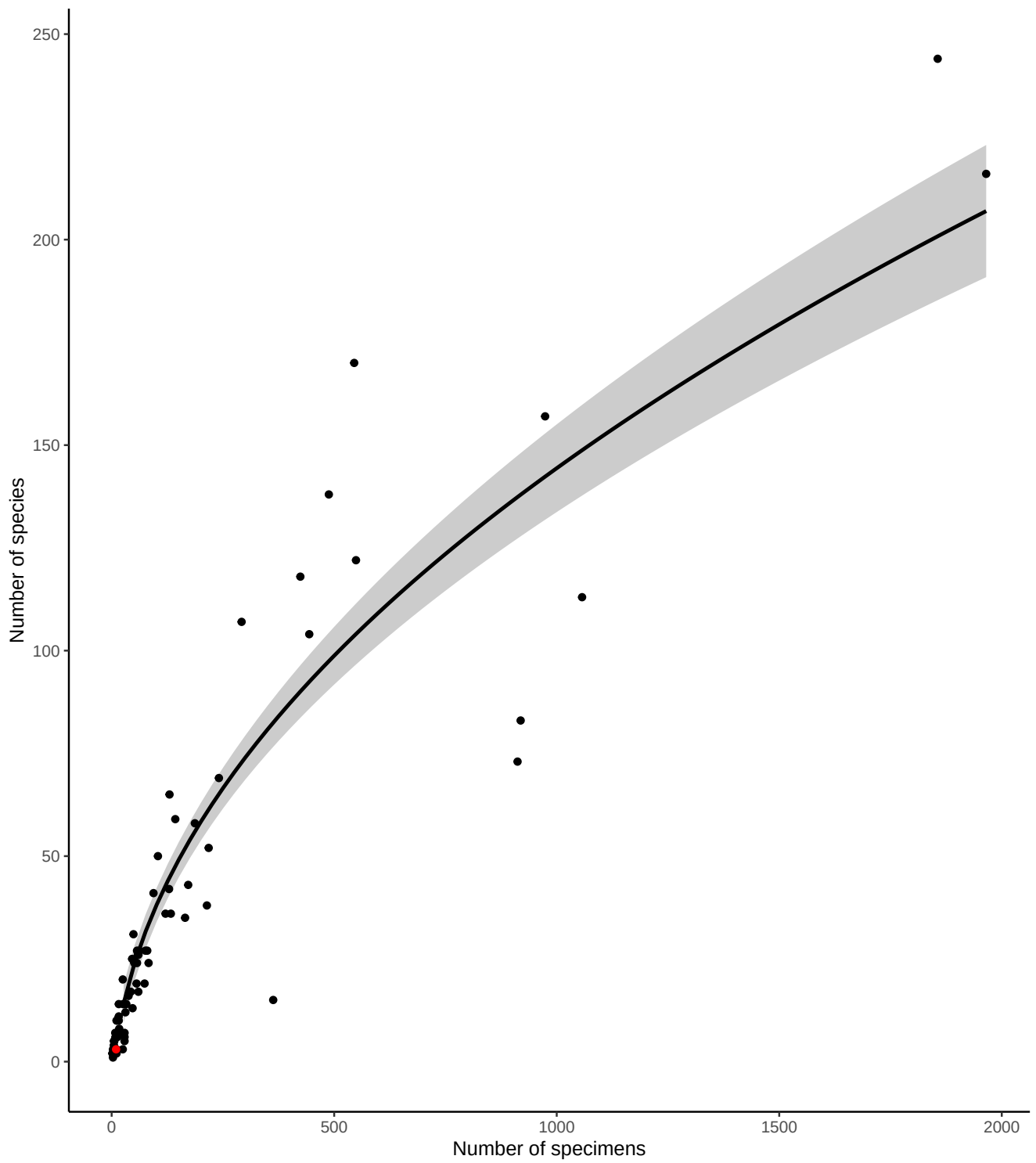


Figure 5: Number of bee specimens and unique bee species caught by all volunteers, with your effort shown in red. This graph should give you an idea of how many specimens you would need to catch to begin seeing rarer bee species.

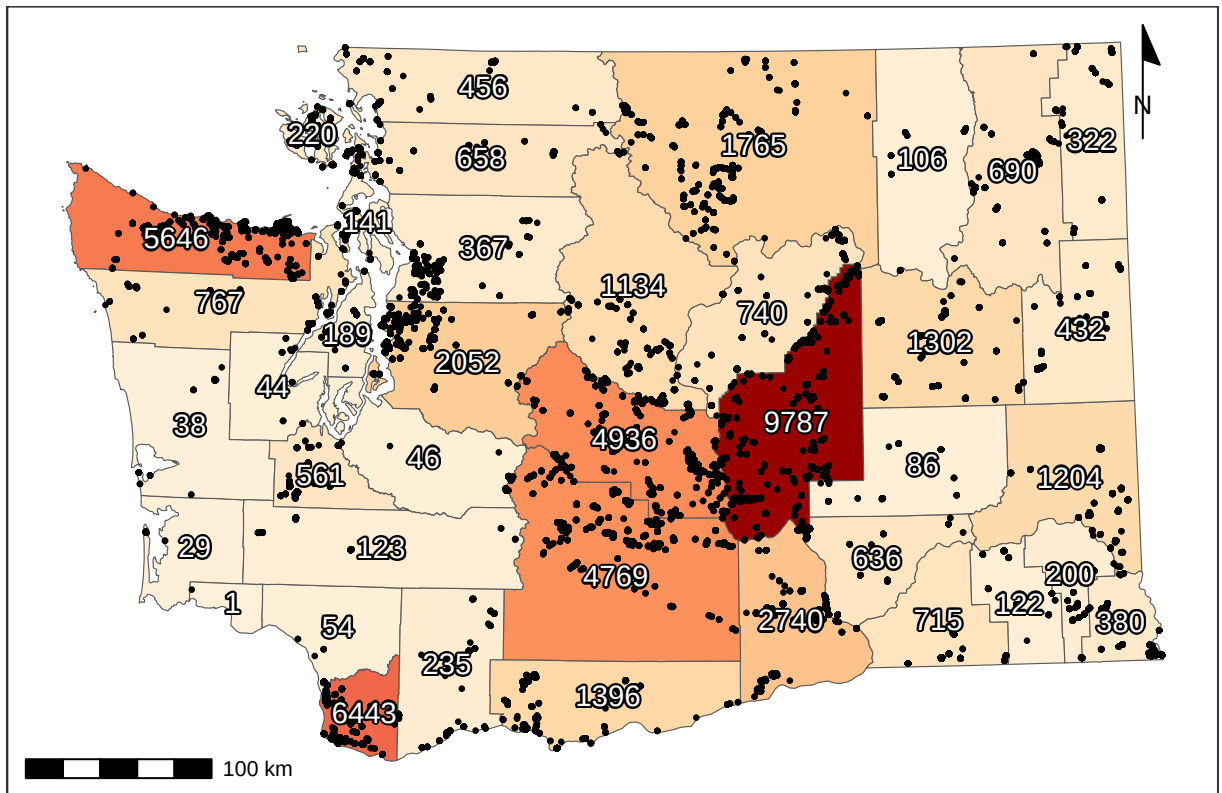


Figure 6: Total specimens caught per county, along with catch location of each specimen (black dots). For genus- and species-specific information for each county, see Tables 3 and 4.

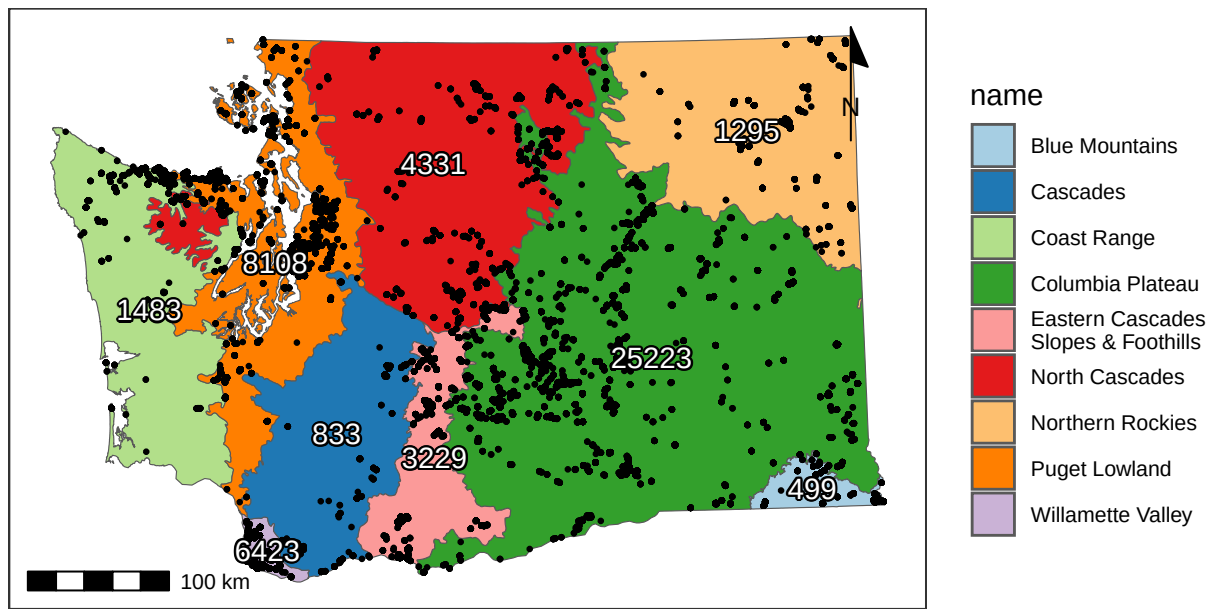


Figure 7: Total catches per ecoregion (Level III), along with catch location of each specimen (black dots).

4 Flight Phenology

4.1 West of (and including) the Cascade Mountains

Most bees (90%) were caught between April 16 and September 01, but the peak of season (50% of specimens) was from June 11 to July 27.

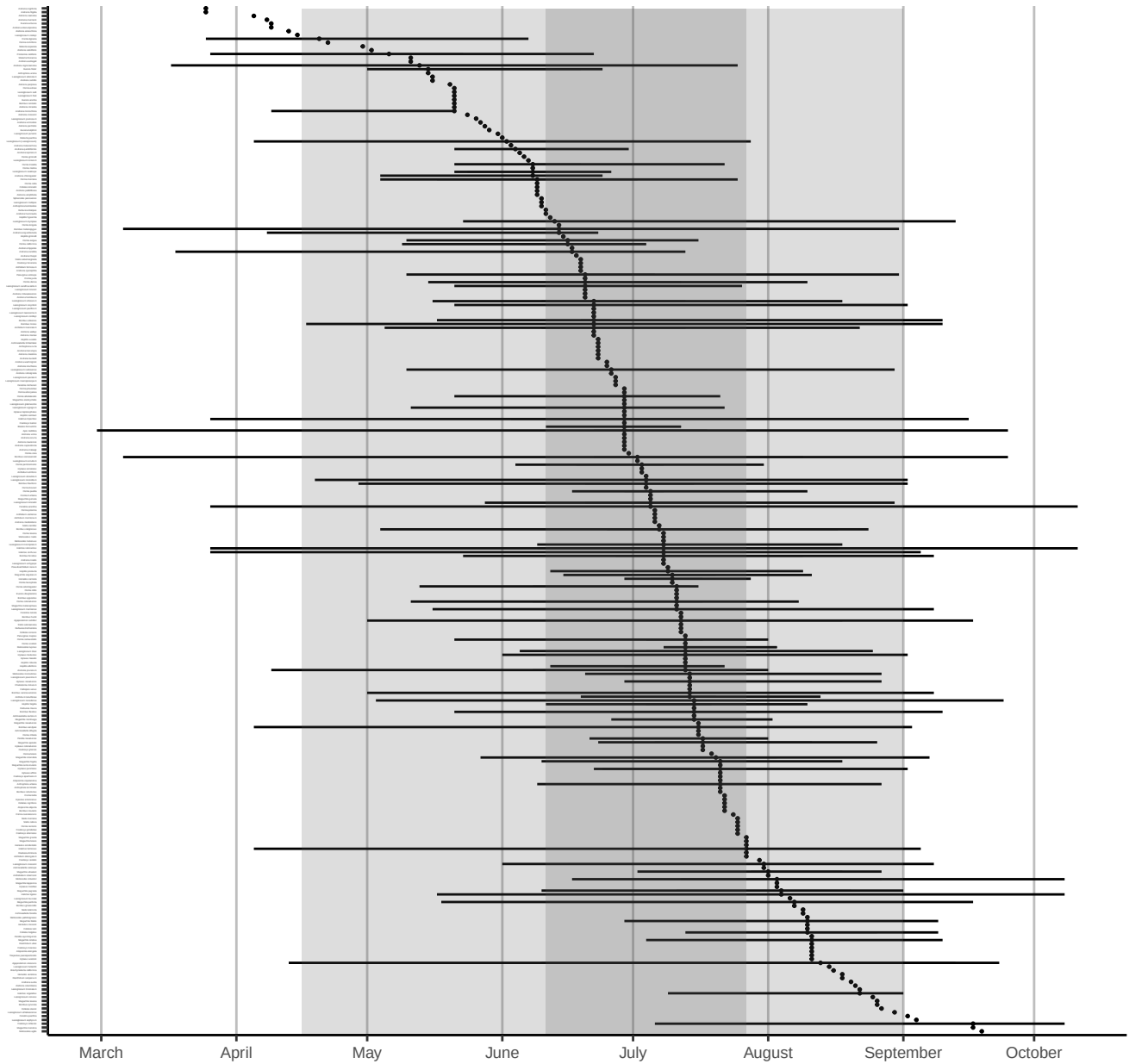


Figure 8: Phenology plot for all bee species caught in or West of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

4.2 East of the Cascade Mountains

Most bees (90%) were caught between April 14 and September 18, but the peak of season (50% of specimens) was from May 15 to July 16.

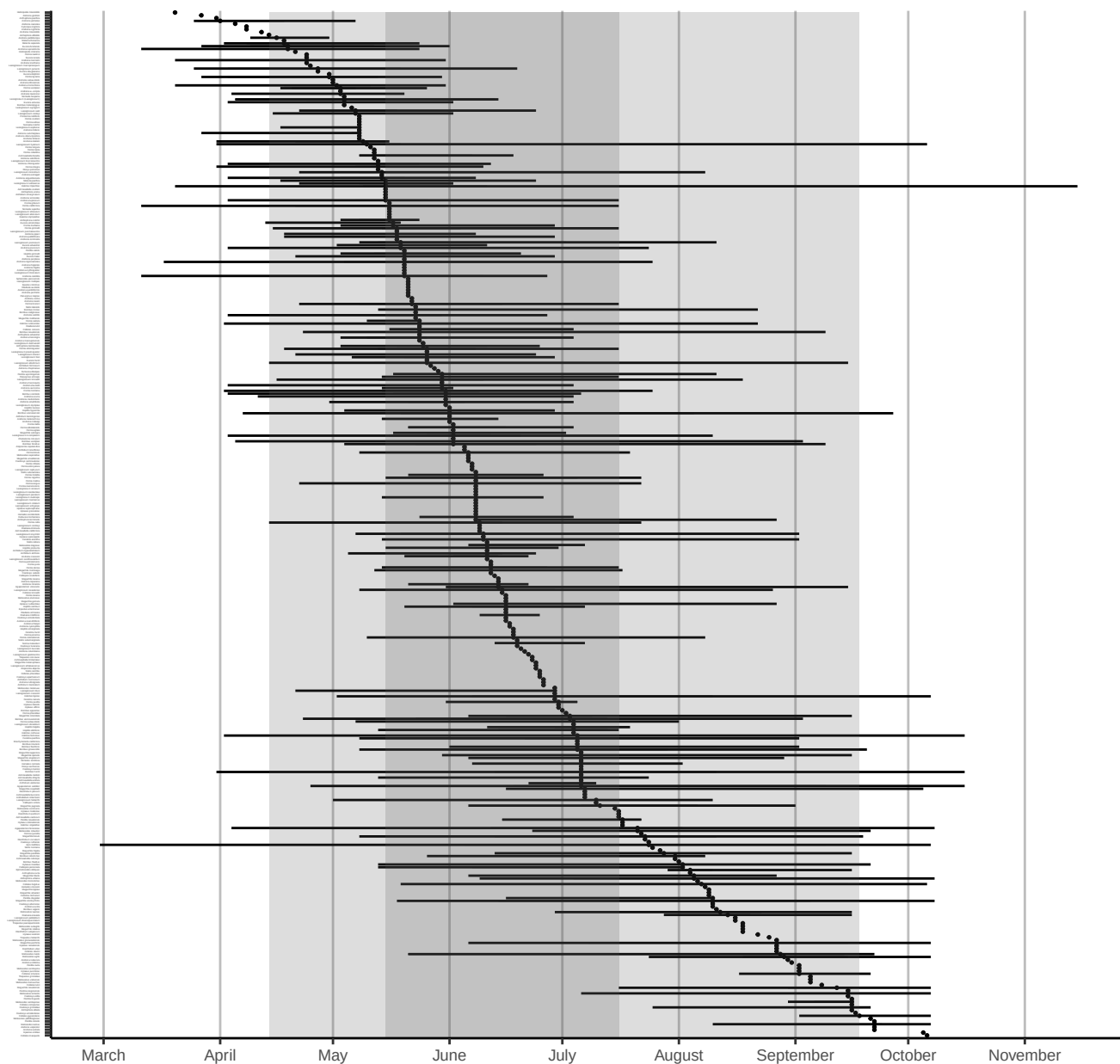


Figure 9: Phenology plot for all bee species caught east of the Cascade Mountains, sorted by median abundance times. Percentiles of overall emergence times (50th & 90th) are shown in grey shaded regions. Date ranges for each species (minimum, first, second, third quartiles, and maximum) are shown only for species with >10 specimens.

5 Plant genera

Volunteers collected specimens from a total of 371 unique flower genera, with most volunteers sampling from 12 flower genera (median value). The **Flower Power Kudos** (most sampled flower genera) goes to Karla Salp, Karen Wright, and Nancy Carlson, who collected bees from a total of 170, 155, and 118 genera of flowers. Well done!

The flower genera that had the most specimens caught on them were *Grindelia*, *Ericameria*, and *Lomatium*, which yielded a total of 1095, 731, and 717 specimens. The flower genera that were popular with the most species of bees were *Erigeron*, *Phacelia*, and *Cirsium*, hosting a total of 101, 98, and 91 unique bee species. See Tables 1 and 2 for more details.

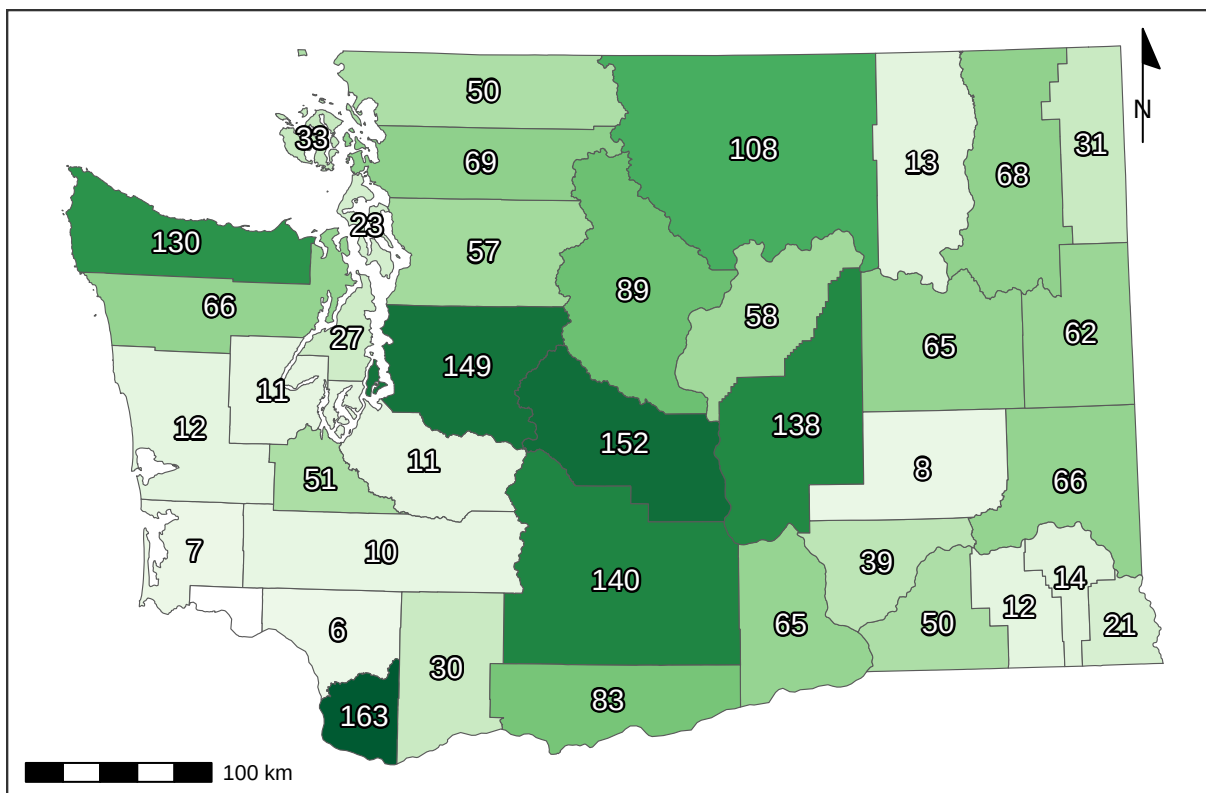


Figure 10: Recorded number of flower genera per county.

Table 1: Number of bee specimens collected from each plant genus. Plants with few records are great targets for future sampling.

Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count
<i>Grindelia</i>	1095	<i>Amsinckia</i>	97	<i>Delphinium</i>	35	<i>Angelica</i>	18	<i>Armeria</i>	8	<i>Anthemis</i>	4	<i>Abelia</i>	2
<i>Ericameria</i>	731	<i>Berberis</i>	97	<i>Malva</i>	35	<i>Acmispon</i>	17	<i>Coriandrum</i>	8	<i>Artemisia</i>	3	<i>Androsace</i>	2
<i>Lomatium</i>	717	<i>Dieteria</i>	94	<i>Nemophila</i>	34	<i>Euphorbia</i>	17	<i>Cuphea</i>	8	<i>Asparagus</i>	3	<i>Arenaria</i>	2
<i>Hypochaeris</i>	655	<i>Eschscholzia</i>	94	<i>Syringa</i>	34	<i>Cacaliopsis</i>	16	<i>Cytisus</i>	8	<i>Barbarea</i>	3	<i>Cardamine</i>	2
<i>Phacelia</i>	624	<i>Fragaria</i>	93	<i>Drymocalis</i>	33	<i>Lewisia</i>	16	<i>Dipsacus</i>	8	<i>Dimorphotheca</i>	3	<i>Corydalis</i>	2
<i>Helianthus</i>	490	<i>Origanum</i>	91	<i>Descurainia</i>	32	<i>Campanula</i>	15	<i>Downingia</i>	8	<i>Eremogone</i>	3	<i>Cota</i>	2
<i>Balsamorhiza</i>	471	<i>Clarkia</i>	89	<i>Malus</i>	32	<i>Draba</i>	15	<i>Hesperochiron</i>	8	<i>Galium</i>	3	<i>Crocasmia</i>	2
<i>Cirsium</i>	450	<i>Medicago</i>	88	<i>Crataegus</i>	31	<i>Hackelia</i>	15	<i>Aquilegia</i>	7	<i>Gayophytum</i>	3	<i>Crocus</i>	2
<i>Erigeron</i>	446	<i>Ceanothus</i>	85	<i>Doellingeria</i>	31	<i>Heterotheca</i>	15	<i>Erica</i>	7	<i>Ceanothus</i>	3	<i>Achlys</i>	1
<i>Solidago</i>	429	<i>Daucus</i>	83	<i>Myosotis</i>	31	<i>Lonicera</i>	15	<i>Fritillaria</i>	7	<i>Hibiscus</i>	3	<i>Adelphia</i>	1
<i>Penstemon</i>	414	<i>Sphaeralcea</i>	82	<i>Tunacetum</i>	31	<i>Oenothera</i>	15	<i>Viburnum</i>	7	<i>Impatiens</i>	3	<i>Albizia</i>	1
<i>Leucanthemum</i>	413	<i>Lathyrus</i>	78	<i>Bellis</i>	30	<i>Perideridia</i>	15	<i>Amelanchier</i>	6	<i>Lapsana</i>	3	<i>Anthozanthum</i>	1
<i>Sisymbrium</i>	408	<i>Pediocactus</i>	78	<i>Cleome</i>	30	<i>Ratibida</i>	15	<i>Anchusa</i>	6	<i>Liatris</i>	3	<i>Antirrhinum</i>	1
<i>Eriogonum</i>	381	<i>Epilobium</i>	77	<i>Hieracium</i>	30	<i>Antennaria</i>	14	<i>Cynara</i>	6	<i>Lobularia</i>	3	<i>Bellardia</i>	1
<i>Lupinus</i>	372	<i>Rudbeckia</i>	77	<i>Rorippa</i>	30	<i>Crocidium</i>	14	<i>Cynoglossum</i>	6	<i>Nothochelone</i>	3	<i>Boechera</i>	1
<i>Potentilla</i>	320	<i>Monardella</i>	74	<i>Cerastium</i>	29	<i>Frangula</i>	14	<i>Limnanthes</i>	6	<i>Oenanthe</i>	3	<i>Buglossoides</i>	1
<i>Rubus</i>	285	<i>Physocarpus</i>	71	<i>Collinsia</i>	29	<i>Hydrophyllum</i>	14	<i>Luetkea</i>	6	<i>Oreocarya</i>	3	<i>Calystegia</i>	1
<i>Rosa</i>	269	<i>Camassia</i>	70	<i>Lactuca</i>	29	<i>Plantago</i>	14	<i>Microseris</i>	6	<i>Physaria</i>	3	<i>Capsella</i>	1
<i>Symphoricarpos</i>	265	<i>Spiraea</i>	70	<i>Olaynia</i>	29	<i>Borago</i>	13	<i>Nigella</i>	6	<i>Pieris</i>	3	<i>Carduus</i>	1
<i>Taraxacum</i>	258	<i>Calochortus</i>	69	<i>Clematis</i>	28	<i>Heleum</i>	13	<i>Onopordum</i>	6	<i>Primula</i>	3	<i>Cercis</i>	1
<i>Madia</i>	256	<i>Nepeta</i>	68	<i>Acer</i>	27	<i>Bistorta</i>	12	<i>Phoenixaulis</i>	6	<i>Pseudognaphalium</i>	3	<i>Chamaecrista</i>	1
<i>Holodiscus</i>	247	<i>Coreopsis</i>	67	<i>Canadanthus</i>	27	<i>Cucurbita</i>	12	<i>Robinia</i>	6	<i>Sagittaria</i>	3	<i>Comandra</i>	1
<i>Achillea</i>	246	<i>Agastache</i>	66	<i>Sonchus</i>	27	<i>Heuchera</i>	12	<i>Ageratina</i>	5	<i>Saponaria</i>	3	<i>Cotoneaster</i>	1
<i>Asclepias</i>	228	<i>Hypericum</i>	60	<i>Sphaerophysa</i>	26	<i>Persicaria</i>	12	<i>Ajuga</i>	5	<i>Silene</i>	3	<i>Cryptantha</i>	1
<i>Lotus</i>	224	<i>Convolvulus</i>	59	<i>Linum</i>	25	<i>Spergularia</i>	12	<i>Ammi</i>	5	<i>Tellima</i>	3	<i>Cucumis</i>	1
<i>Trifolium</i>	218	<i>Cichorium</i>	55	<i>Wyethia</i>	25	<i>Vitex</i>	12	<i>Arctostaphylos</i>	5	<i>Thalictrum</i>	3	<i>Dicentra</i>	1
<i>Crepis</i>	217	<i>Veronica</i>	55	<i>Aruncus</i>	24	<i>Centaurium</i>	11	<i>Bassia</i>	5	<i>Thermopsis</i>	3	<i>Echinops</i>	1
<i>Brassica</i>	213	<i>Arnica</i>	54	<i>Chondrilla</i>	24	<i>Chorispura</i>	11	<i>Cymopterus</i>	5	<i>Visnaga</i>	3	<i>Erythronium</i>	1
<i>Centauria</i>	210	<i>Digitalis</i>	54	<i>Lithospermum</i>	24	<i>Cosmos</i>	11	<i>Helopsis</i>	5	<i>Dalea</i>	2	<i>Gnaphalium</i>	1
<i>Prunus</i>	202	<i>Linaria</i>	53	<i>Anethum</i>	23	<i>Dahlia</i>	11	<i>Hydrangea</i>	5	<i>Eurybia</i>	2	<i>Holcus</i>	1
<i>Salix</i>	200	<i>Chaenactis</i>	52	<i>Purshia</i>	23	<i>Micranthes</i>	11	<i>Melissa</i>	5	<i>Fagopyrum</i>	2	<i>Hylotelephium</i>	1
<i>Ribes</i>	188	<i>Nothocalais</i>	51	<i>Hieracium</i>	22	<i>Pseudotsuga</i>	11	<i>Mycelis</i>	5	<i>Fothergilla</i>	2	<i>Hymenopappus</i>	1
<i>Ranunculus</i>	184	<i>Sedum</i>	51	<i>Mentzelia</i>	22	<i>Pyrocampa</i>	11	<i>Narcissus</i>	5	<i>Hesperis</i>	2	<i>Hyssopus</i>	1
<i>Anaphalis</i>	179	<i>Euthamia</i>	47	<i>Thelypodium</i>	22	<i>Valerianella</i>	11	<i>Petasites</i>	5	<i>Ionactis</i>	2	<i>Ilex</i>	1
<i>Senecio</i>	176	<i>Cornus</i>	44	<i>Triteleia</i>	22	<i>Asperugo</i>	10	<i>Phedimus</i>	5	<i>Kalmia</i>	2	<i>Ipomopsis</i>	1
<i>Chrysothamnus</i>	175	<i>Helianthella</i>	44	<i>Papaver</i>	21	<i>Berteroa</i>	10	<i>Sorbus</i>	5	<i>Ladecania</i>	2	<i>Isocoma</i>	1
<i>Melilotus</i>	167	<i>Lavandula</i>	44	<i>Polemonium</i>	21	<i>Erodium</i>	10	<i>Argentina</i>	4	<i>Ligusticum</i>	2	<i>Lamprocapnos</i>	1
<i>Gaillardia</i>	154	<i>Mentha</i>	44	<i>Erythranthe</i>	20	<i>Lamium</i>	10	<i>Cistus</i>	4	<i>Lycium</i>	2	<i>Lomelosia</i>	1
<i>Chamaenerion</i>	147	<i>Plagiobothrys</i>	44	<i>Frasera</i>	20	<i>Lygodesmia</i>	10	<i>Dichelostemma</i>	4	<i>Mertensia</i>	2	<i>Lycopus</i>	1
<i>Philadelphus</i>	146	<i>Plectritis</i>	43	<i>Iris</i>	20	<i>Marrubium</i>	10	<i>Echinacea</i>	4	<i>Montia</i>	2	<i>Matricaria</i>	1
<i>Albium</i>	144	<i>Pastinaca</i>	42	<i>Oemeria</i>	20	<i>Monarda</i>	10	<i>Elaeagnus</i>	4	<i>Nasturtium</i>	2	<i>Muscari</i>	1
<i>Lepidium</i>	134	<i>Rhododendron</i>	42	<i>Phlox</i>	20	<i>Solanum</i>	10	<i>Eutrochium</i>	4	<i>Noccea</i>	2	<i>Nicotiana</i>	1
<i>Symphoricarpos</i>	132	<i>Vaccinium</i>	42	<i>Rhus</i>	20	<i>Tragopogon</i>	10	<i>Glycyrrhiza</i>	4	<i>Osmia</i>	2	<i>Oxalis</i>	1
<i>Vicia</i>	120	<i>Lithophragma</i>	40	<i>Toxicoscordion</i>	20	<i>Fallopia</i>	9	<i>Iberis</i>	4	<i>Ozomelis</i>	2	<i>Phalaris</i>	1
<i>Cilia</i>	119	<i>Erysimum</i>	39	<i>Abronia</i>	19	<i>Pyrrus</i>	9	<i>Ivesia</i>	4	<i>Pedicularis</i>	2	<i>Pisum</i>	1
<i>Salvia</i>	117	<i>Nestodus</i>	39	<i>Anthriscus</i>	19	<i>Rhaphiolepis</i>	9	<i>Machaeranthera</i>	4	<i>Petroselinum</i>	2	<i>Polygonum</i>	1
<i>Jacobaea</i>	116	<i>Calendula</i>	38	<i>Castilleja</i>	19	<i>Tripleurospermum</i>	9	<i>Microsteris</i>	4	<i>Pilosella</i>	2	<i>Salsola</i>	1
<i>Eriophyllum</i>	111	<i>Dasiphora</i>	38	<i>Gaultheria</i>	19	<i>Leontodon</i>	8	<i>Mutarda</i>	4	<i>Poa</i>	2	<i>Sambucus</i>	1
<i>Astragalus</i>	104	<i>Sidalcea</i>	38	<i>Thymus</i>	19	<i>Levistica</i>	8	<i>Opuntia</i>	4	<i>Pyraantha</i>	2	<i>Santolina</i>	1
<i>Apocynum</i>	102	<i>Cakile</i>	37	<i>Collomia</i>	18	<i>Paeonia</i>	8	<i>Petrosedum</i>	4	<i>Satureja</i>	2	<i>Sisyrinchium</i>	1
<i>Geranium</i>	99	<i>Claytonia</i>	37	<i>Oxytropis</i>	18	<i>Scrophularia</i>	8	<i>Stellaria</i>	4	<i>Symphytum</i>	2	<i>Urtica</i>	1
<i>Ilamna</i>	99	<i>Prunella</i>	35	<i>Stephanomeria</i>	18	<i>Stachys</i>	8	<i>Tolmiea</i>	4	<i>Viola</i>	2	<i>Veratrum</i>	1
<i>Lythrum</i>	97	<i>Rhaponticum</i>	35	<i>Tagetes</i>	18	<i>Valeriana</i>	8	<i>Verbena</i>	4	<i>Zinnia</i>	2	<i>Xerophyllum</i>	1

Table 2: Number of bee species collected from each plant genus

Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count	Genus	Count
<i>Erigeron</i>	101	<i>Coreopsis</i>	31	<i>Bellis</i>	15	<i>Antennaria</i>	8	<i>Ageratina</i>	4	<i>Argentina</i>	2	<i>Abelia</i>	1
<i>Phacelia</i>	98	<i>Agastache</i>	29	<i>Crataegus</i>	14	<i>Aruncus</i>	8	<i>Ajuga</i>	4	<i>Artemisia</i>	2	<i>Achlys</i>	1
<i>Cirsium</i>	91	<i>Eschscholzia</i>	29	<i>Hieracium</i>	14	<i>Berteroa</i>	8	<i>Ammi</i>	4	<i>Borago</i>	2	<i>Adelphia</i>	1
<i>Grindelia</i>	91	<i>Monardella</i>	29	<i>Hieracium</i>	14	<i>Cacaliopsis</i>	8	<i>Cistus</i>	4	<i>Cardamine</i>	2	<i>Albizia</i>	1
<i>Eriogonum</i>	86	<i>Berberis</i>	28	<i>Malus</i>	14	<i>Canadanthus</i>	8	<i>Cosmos</i>	4	<i>Cota</i>	2	<i>Androsace</i>	1
<i>Penstemon</i>	85	<i>Chrysothamnus</i>	28	<i>Malva</i>	14	<i>Cleome</i>	8	<i>Cuphea</i>	4	<i>Crocosmia</i>	2	<i>Anthozanthum</i>	1
<i>Solidago</i>	82	<i>Convolvulus</i>	28	<i>Tanacetum</i>	14	<i>Erysimum</i>	8	<i>Cymopterus</i>	4	<i>Dalea</i>	2	<i>Antirrhinum</i>	1
<i>Crepis</i>	75	<i>Dieteria</i>	28	<i>Lithospermum</i>	13	<i>Heuchera</i>	8	<i>Dahlia</i>	4	<i>Eurybia</i>	2	<i>Arenaria</i>	1
<i>Hypochaeris</i>	75	<i>Fragaria</i>	28	<i>Parashia</i>	13	<i>Hydrophyllum</i>	8	<i>Dichlostenma</i>	4	<i>Galium</i>	2	<i>Bellardia</i>	1
<i>Lupinus</i>	75	<i>Calochortus</i>	27	<i>Sonchus</i>	13	<i>Anethum</i>	7	<i>Echinacea</i>	4	<i>Geum</i>	2	<i>Boechera</i>	1
<i>Leucanthemum</i>	69	<i>Chaenactis</i>	27	<i>Collomia</i>	12	<i>Anthriscus</i>	7	<i>Eutrochium</i>	4	<i>Glycyrrhiza</i>	2	<i>Buglossoides</i>	1
<i>Rosa</i>	69	<i>Lathyrus</i>	27	<i>Doellingeria</i>	12	<i>Campanula</i>	7	<i>Frangula</i>	4	<i>Hesperis</i>	2	<i>Calystegia</i>	1
<i>Potentilla</i>	67	<i>Madia</i>	26	<i>Hackelia</i>	12	<i>Centaurium</i>	7	<i>Fritillaria</i>	4	<i>Iberis</i>	2	<i>Capsella</i>	1
<i>Sisymbrium</i>	64	<i>Medicago</i>	26	<i>Lactuca</i>	12	<i>Helenium</i>	7	<i>Heliopsis</i>	4	<i>Impatiens</i>	2	<i>Carduus</i>	1
<i>Achillea</i>	63	<i>Hypericum</i>	25	<i>Plantago</i>	12	<i>Heterotheca</i>	7	<i>Hydrangea</i>	4	<i>Ionactis</i>	2	<i>Cercis</i>	1
<i>Ericameria</i>	63	<i>Rudbeckia</i>	25	<i>Syringa</i>	12	<i>Micranthes</i>	7	<i>Ivesia</i>	4	<i>Kalmia</i>	2	<i>Chamaecrista</i>	1
<i>Lomatium</i>	63	<i>Arnica</i>	24	<i>Triteleia</i>	12	<i>Perideridia</i>	7	<i>Machaeranthera</i>	4	<i>Levisticum</i>	2	<i>Comandra</i>	1
<i>Balsamorhiza</i>	62	<i>Lythrum</i>	24	<i>Vaccinium</i>	12	<i>Rorippa</i>	7	<i>Microsteris</i>	4	<i>Liatris</i>	2	<i>Corydalis</i>	1
<i>Rubus</i>	62	<i>Nepeta</i>	24	<i>Wyethia</i>	12	<i>Stephanomeria</i>	7	<i>Mycelis</i>	4	<i>Ligusticum</i>	2	<i>Cotoneaster</i>	1
<i>Centauria</i>	60	<i>Cichorium</i>	23	<i>Collinsia</i>	11	<i>Abronia</i>	6	<i>Narcissus</i>	4	<i>Limnanthes</i>	2	<i>Crocus</i>	1
<i>Chamaenerion</i>	59	<i>Clematis</i>	23	<i>Lonicera</i>	11	<i>Amelanchier</i>	6	<i>Oenleria</i>	4	<i>Labularia</i>	2	<i>Cryptantha</i>	1
<i>Allium</i>	58	<i>Epilobium</i>	23	<i>Mentzelia</i>	11	<i>Aquilegia</i>	6	<i>Onopordum</i>	4	<i>Lycium</i>	2	<i>Cucumis</i>	1
<i>Senecio</i>	51	<i>Claytonia</i>	22	<i>Olsynium</i>	11	<i>Armeria</i>	6	<i>Phedimus</i>	4	<i>Montia</i>	2	<i>Dicentra</i>	1
<i>Helianthus</i>	50	<i>Euthamia</i>	22	<i>Sphaerophysa</i>	11	<i>Castilleja</i>	6	<i>Stachys</i>	4	<i>Nasturtium</i>	2	<i>Downingia</i>	1
<i>Holodiscus</i>	50	<i>Ilama</i>	21	<i>Thelypodium</i>	11	<i>Crocidium</i>	6	<i>Toxicoscordion</i>	4	<i>Nigella</i>	2	<i>Echinops</i>	1
<i>Melilotus</i>	50	<i>Linaria</i>	21	<i>Thymus</i>	11	<i>Cucurbita</i>	6	<i>Anthemis</i>	3	<i>Nothochelone</i>	2	<i>Erythronium</i>	1
<i>Prunus</i>	50	<i>Nothocalais</i>	21	<i>Acer</i>	10	<i>Erica</i>	6	<i>Arctostaphylos</i>	3	<i>Opuntia</i>	2	<i>Fagopyrum</i>	1
<i>Trifolium</i>	50	<i>Sedum</i>	21	<i>Chondrilla</i>	10	<i>Fullopia</i>	6	<i>Asparagus</i>	3	<i>Osmia</i>	2	<i>Fothergilla</i>	1
<i>Lotus</i>	49	<i>Philadelphus</i>	20	<i>Descurainia</i>	10	<i>Hesperochiron</i>	6	<i>Barbarea</i>	3	<i>Oxytropis</i>	2	<i>Gnaphalium</i>	1
<i>Taraxacum</i>	49	<i>Physocarpus</i>	20	<i>Digitalis</i>	10	<i>Leontodon</i>	6	<i>Bassia</i>	3	<i>Pedicularis</i>	2	<i>Holcus</i>	1
<i>Anaphalis</i>	45	<i>Sphaeralcea</i>	20	<i>Euphorbia</i>	10	<i>Monarda</i>	6	<i>Cynara</i>	3	<i>Petroselinum</i>	2	<i>Hylotelephium</i>	1
<i>Gaillardia</i>	45	<i>Camassia</i>	19	<i>Lewisia</i>	10	<i>Persicaria</i>	6	<i>Dimorphotheca</i>	3	<i>Physaria</i>	2	<i>Hymenopappus</i>	1
<i>Salvia</i>	45	<i>Cornus</i>	19	<i>Oenothera</i>	10	<i>Rhaphiolepis</i>	6	<i>Elaeagnus</i>	3	<i>Pieris</i>	2	<i>Hyssopus</i>	1
<i>Brassica</i>	43	<i>Plectritis</i>	19	<i>Plagiobothrys</i>	10	<i>Spergularia</i>	6	<i>Eremogone</i>	3	<i>Pyraecantha</i>	2	<i>Ilex</i>	1
<i>Ranunculus</i>	43	<i>Veronica</i>	19	<i>Polemonium</i>	10	<i>Tripleurospermum</i>	6	<i>Gayophytum</i>	3	<i>Stellaria</i>	2	<i>Ipomopsis</i>	1
<i>Symphotrichum</i>	43	<i>Daucus</i>	18	<i>Pseudotsuga</i>	10	<i>Valerianella</i>	6	<i>Hibiscus</i>	3	<i>Symphytum</i>	2	<i>Isocoma</i>	1
<i>Vicia</i>	43	<i>Delphinium</i>	18	<i>Acmispon</i>	9	<i>Anchusa</i>	5	<i>Lamium</i>	3	<i>Thermopsis</i>	2	<i>Ladecania</i>	1
<i>Lepidium</i>	42	<i>Drymocallis</i>	18	<i>Angelica</i>	9	<i>Asperugo</i>	5	<i>Lapsana</i>	3	<i>Viola</i>	2	<i>Lamprocapnos</i>	1
<i>Salix</i>	38	<i>Mentha</i>	18	<i>Bistorta</i>	9	<i>Coriandrum</i>	5	<i>Luetkea</i>	3	<i>Polygonum</i>	1	<i>Lomelosia</i>	1
<i>Symphoricarpos</i>	38	<i>Nastotus</i>	18	<i>Cerastium</i>	9	<i>Cynoglossum</i>	5	<i>Mutarda</i>	3	<i>Salsola</i>	1	<i>Lycopus</i>	1
<i>Asclepias</i>	37	<i>Prunella</i>	18	<i>Chorispora</i>	9	<i>Cytisus</i>	5	<i>Oenanthe</i>	3	<i>Sambucus</i>	1	<i>Matricaria</i>	1
<i>Amsinckia</i>	35	<i>Erythranthe</i>	17	<i>Erodium</i>	9	<i>Dipsacus</i>	5	<i>Oreocarya</i>	3	<i>Santolina</i>	1	<i>Melissa</i>	1
<i>Apocynum</i>	35	<i>Helianthella</i>	17	<i>Frasera</i>	9	<i>Draba</i>	5	<i>Phoenicautis</i>	3	<i>Satureja</i>	1	<i>Mertensia</i>	1
<i>Eriophyllum</i>	35	<i>Linum</i>	17	<i>Iris</i>	9	<i>Gaultheria</i>	5	<i>Primula</i>	3	<i>Sisyrinchium</i>	1	<i>Muscari</i>	1
<i>Geranium</i>	34	<i>Rhaponticum</i>	17	<i>Nemophila</i>	9	<i>Microseris</i>	5	<i>Pseudognaphalium</i>	3	<i>Tellima</i>	1	<i>Nicotiana</i>	1
<i>Jacobaea</i>	34	<i>Cakile</i>	16	<i>Pastinaca</i>	9	<i>Paeonia</i>	5	<i>Pyrus</i>	3	<i>Thalictrum</i>	1	<i>Noccea</i>	1
<i>Ribes</i>	34	<i>Pedicularis</i>	16	<i>Lithophragma</i>	8	<i>Petasites</i>	5	<i>Robinia</i>	3	<i>Tobmicea</i>	1	<i>Ozalis</i>	1
<i>Ceanothus</i>	33	<i>Sidalcea</i>	16	<i>Lygodesmia</i>	8	<i>Phlox</i>	5	<i>Sagittaria</i>	3	<i>Urtica</i>	1	<i>Ozomelis</i>	1
<i>Clarkia</i>	33	<i>Calendula</i>	15	<i>Marrubium</i>	8	<i>Pyrrocoma</i>	5	<i>Saponaria</i>	3	<i>Veratrum</i>	1	<i>Petrosedum</i>	1
<i>Origanum</i>	33	<i>Dasiphora</i>	15	<i>Papaver</i>	8	<i>Solanum</i>	5	<i>Scrophularia</i>	3	<i>Verbena</i>	1	<i>Phalaris</i>	1
<i>Spiraea</i>	33	<i>Lavandula</i>	15	<i>Ratibida</i>	8	<i>Tragopogon</i>	5	<i>Silene</i>	3	<i>Visnaga</i>	1	<i>Pilosella</i>	1
<i>Astragalus</i>	32	<i>Myosotis</i>	15	<i>Rhus</i>	8	<i>Valeriana</i>	4	<i>Sorbus</i>	3	<i>Xerophyllum</i>	1	<i>Pisum</i>	1
<i>Gilia</i>	32	<i>Rhododendron</i>	15	<i>Tagetes</i>	8	<i>Viburnum</i>	4	<i>Vitex</i>	3	<i>Zinnia</i>	1	<i>Poa</i>	1

	Adams	Asotin	Benton	Chelan	Cllallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Island	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Skagit	Skamania	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whitcom	Whitman	Yakima	TOTAL	
Andrena vicinaoides	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	1	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	4	12	
Andrena vulpcolor	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
Andrena w-scripta	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	3	
Andrena wallgii	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
Andrena washingtoni	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
Anthididum robertsoni	0	0	0	0	0	0	0	0	0	0	2	0	5	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	11
Anthididum atrifrons	0	0	0	0	0	0	0	0	0	0	2	3	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	19
Anthididum banningsense	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	7
Anthididum clypeodentatum	0	0	0	0	0	0	0	0	3	0	0	0	4	0	0	0	0	0	0	0	0	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10
Anthididum emarginatum	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	5
Anthididum formosum	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	2
Anthididum manicatum	0	0	2	0	4	9	0	0	0	0	1	0	1	1	0	1	4	4	0	0	0	0	0	0	0	0	0	1	0	0	1	3	0	1	0	0	0	0	0	1	34
Anthididum mormonum	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0	0	4	0	0	0	0	0	0	0	0	2	1	0	0	0	0	0	4	15
Anthididum oblongatum	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	23	
Anthididum tenuiflorae	0	0	0	1	3	0	0	0	0	0	0	0	1	0	0	0	0	0	7	0	0	0	0	1	0	0															

Table 4: Number of bee specimens from each county, by species (continued)

	Adams	Asotin	Benton	Chelan	Clallam	Clark	Columbia	Cowlitz	Douglas	Ferry	Franklin	Garfield	Grant	Grays Harbor	Inland	Jefferson	King	Kitsap	Kittitas	Klickitat	Lewis	Lincoln	Mason	Okanogan	Pacific	Pend Oreille	Pierce	San Juan	Shagit	Shanawia	Snohomish	Spokane	Stevens	Thurston	Wahkiakum	Walla Walla	Whatcom	Whitman	Yakima	TOTAL	
<i>Osmia cobaltina</i>	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	3	
<i>Osmia coloradensis</i>	0	0	0	9	1	0	0	0	1	1	0	1	1	0	0	1	1	2	11	0	0	1	0	7	0	4	0	2	1	0	0	3	0	0	0	1	1	3	15	67	
<i>Osmia cornifrons</i>	0	0	0	0	0	6	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	9	
<i>Osmia cyanea</i>	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Osmia densa</i>	0	0	0	5	1	0	0	0	0	1	0	0	0	0	0	0	0	8	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	23
<i>Osmia ednae</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	
<i>Osmia exigua</i>	0	0	1	2	1	1	0	0	0	0	0	2	3	0	0	2	0	0	10	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	11	35	
<i>Osmia gliarum</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia grinnelli</i>	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	6	24
<i>Osmia integra</i>	0	0	0	0	0	0	0	0	0	0	4	0	29	0	0	0	0	0	8	0	0	0	0	6	0	0	0	0	0	0	0	0	0	0	0	0	0	1	10	58	
<i>Osmia inurbana</i>	0	0	0	0	0	0	0	1	1	2	0	1	1	0	0	0	0	0	5	2	0	1	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	25	
<i>Osmia iridis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia juxta</i>	0	0	0	2	6	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
<i>Osmia lacta</i>	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia lignaria</i>	0	0	0	1	4	6	0	0	0	0	0	0	3	0	0	3	2	0	4	1	0	0	0	3	0	0	0	0	0	6	1	0	0	0	0	0	1	0	5	40	
<i>Osmia longula</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	1	7		
<i>Osmia malina</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia montana</i>	0	0	0	10	0	0	0	0	0	0	0	1	2	0	0	0	0	0	22	9	0	1	0	27	0	2	0	0	0	0	0	0	0	0	0	0	0	1	26	101	
<i>Osmia nemoris</i>	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia nifata</i>	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	5		
<i>Osmia nigroventris</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Osmia odontogaster</i>	0	0	0	1	4	0	0	0	0	0	0	3	0	0	0	0	0	0	16	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	32
<i>Osmia penstemonis</i>	0	0	0	11	1	0	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	3	0	4	4	0	1	1	0	0	0	0	0	0	0	0	4	33	
<i>Osmia phacellae</i>	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	8	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	
<i>Osmia proxima</i>	0	0	0	0	2	2	0	0	0	0	0	0	0	0	0	1	9	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	1	1	19	
<i>Osmia pusilla</i>	0	0	0	5	5	10	0	0	0	0	0	0	0	0	0	0	0	0	27	0	0	0	0	3	0	1	0	0	0	0	0	0	0	0	0	6	0	1	1	59	
<i>Osmia rasilinist</i>	0	0	0	0	0	0	0	0	0	0	0	0	4	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	7	
<i>Osmia regulina</i>	0	0	0	1	0	0	0	0	0	0	5	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	13	0	0	2	25		
<i>Osmia sculleni</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	6	
<i>Osmia sedula</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	12	
<i>Osmia subaustrials</i>	0	0	0	2	0	0	0	0	0	0	0	0	2	0	0	0	0	0	6	0	0	0	4	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	4	19	
<i>Osmia tezana</i>	0	0	0	0	0	4	0	0	0	0	0	0	1	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	11		
<i>Osmia trifoliata</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1		
<i>Osmia tristella</i>	0	0	0	7	6	0	0	0	0	0	0	1	0	0	0	0	5	0	17	0	0	0	1	0	16	0	0	2	0	0	0	0	0	0	0	1	0	0	30	86	
<i>Osmia vandykei</i>	0	0	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	15	
<i>Panurginus atriceps</i>	0	0	0	11	45	9	0	0	0	0	0	2	0	0	0	10	1	0	35	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	8	4	126		
<i>Panurginus inepus</i>	0	0	0	5	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	
<i>Perdita claypolei</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	11	11	
<i>Perdita lingualis</i>	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	2		
<i>Perdita nevadensis</i>	0	0	0	12	0	0	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	24	
<i>Perdita nuda</i>	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Perdita oregonensis</i>	0	0	0	0	0	0	0	0	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	5	
<i>Perdita salicis</i>	0	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Perdita wyomingensis</i>	0	0	0	1	0	0	8	0	0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	13		
<i>Perdita zonalis</i>	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	
<i>Protosmia rubifloris</i>	0	0	0	1	0	6	0	0	0	0	0	0	1	0	0	0	0	1	2	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	16	
<i>Pseudoanthidium nanum</i>	0	0	0	0	0	9	0	0	0	0	0																														

Table 5: Your determination accuracy in 2024.

Taxon
No specimens identified

7 Taxonomic Accuracy, 2024

In 2024, you identified 0 of your 11 specimens to genus level and 0 to species level (see Table 5). In total, volunteers from the Washington Bee Atlas project identified 1.3 % (127) of the 10006 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (see Table 6).

Table 6: Determination accuracy for all volunteers in 2024.

Taxon	Specimens ID-ed	Correct ID	% Correct
Family			
<i>Andrenidae</i>	21	16	76.2
<i>Apidae</i>	52	52	100.0
<i>Colletidae</i>	6	3	50.0
<i>Halictidae</i>	8	6	75.0
<i>Megachilidae</i>	43	41	95.3
<i>TOTAL</i>	130	118	90.8
Genus			
<i>Agapostemon</i>	1	1	100.0
<i>Andrena</i>	5	0	0.0
<i>Anthidium</i>	1	1	100.0
<i>Anthophora</i>	6	6	100.0
<i>Ceratina</i>	7	7	100.0
<i>Coelioxys</i>	2	2	100.0
<i>Colletes</i>	4	3	75.0
<i>Eucera</i>	1	1	100.0
<i>Halictus</i>	2	0	0.0
<i>Hoplitis</i>	4	4	100.0
<i>Hylaeus</i>	2	0	0.0
<i>Lasioglossum</i>	5	4	80.0
<i>Megachile</i>	18	18	100.0
<i>Melissodes</i>	36	33	91.7
<i>Osmia</i>	17	15	88.2
<i>Perdita</i>	16	16	100.0
<i>TOTAL</i>	127	111	87.4
Species			
<i>TOTAL</i>	0	0	NaN

Table 7: Your determination accuracy.

Taxon
No specimens identified

8 Taxonomic Accuracy, All Years

Over your time in the Atlas you identified 0 of your 11 specimens to genus level and 0 to species level (see Table 7). In total, volunteers from the Washington Bee Atlas project identified 0.6 % (127) of the 22418 bee specimens to the level of genus, with an average accuracy of 87.4%. Volunteer identifications to species level did not occur or are not included in the data (See Table 8).

Table 8: Determination accuracy for all volunteers.

Taxon	Specimens ID-ed	Correct ID	% Correct
Family			
<i>Andrenidae</i>	21	16	76.2
<i>Apidae</i>	52	52	100.0
<i>Colletidae</i>	6	3	50.0
<i>Halictidae</i>	8	6	75.0
<i>Megachilidae</i>	43	41	95.3
<i>TOTAL</i>	130	118	90.8
Genus			
<i>Agapostemon</i>	1	1	100.0
<i>Andrena</i>	5	0	0.0
<i>Anthidium</i>	1	1	100.0
<i>Anthophora</i>	6	6	100.0
<i>Ceratina</i>	7	7	100.0
<i>Coelioxys</i>	2	2	100.0
<i>Colletes</i>	4	3	75.0
<i>Eucera</i>	1	1	100.0
<i>Halictus</i>	2	0	0.0
<i>Hoplitis</i>	4	4	100.0
<i>Hylaeus</i>	2	0	0.0
<i>Lasioglossum</i>	5	4	80.0
<i>Megachile</i>	18	18	100.0
<i>Melissodes</i>	36	33	91.7
<i>Osmia</i>	17	15	88.2
<i>Perdita</i>	16	16	100.0
<i>TOTAL</i>	127	111	87.4
Species			
<i>TOTAL</i>	0	0	NaN